

DENIS FEDERICO LIM

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RESEARCH

Representational Adversarial Steganography in Colluding LLMs

March 2026

- Co-authored paper at Apart Research hackathon (co-organized with Redwood Research) on collusive representational steganography as a multi-agent AI control threat model, formalized as a three-player adversarial game.
- Implemented co-evolutionary adversarial training protocol using GPT-2 small with a rank-8 latent MLP channel; achieved near-perfect covert signal transmission while degrading monitor accuracy to ~20% after 15 adversarial rounds.

Chain-of-thought Faithfulness in Small Reasoning Models

November 2025

- Designed and implemented an end-to-end mechanistic interpretability pipeline to evaluate CoT faithfulness in DeepSeek-R1-Distill-Qwen-1.5B using question-flipping methodology
- Cached residual stream activations using TransformerLens hooks; trained linear probes to classify faithfulness from internal activations.

Medical Inhaler Spacer Research — USC Viterbi Senior Design

January 2017 – May 2017

- Collaborated with Children’s Hospital of Los Angeles to design and test pressurized medicinal inhaler spacer. Implemented Digital Image Correlation (DIC) methods with MATLAB to numerically analyze fluid dynamics and optimize device geometry through systematic experimentation.

FELLOWSHIPS & PROGRAMS

Arcadia Alignment (Arcadia Impact) — Automated Alignment Research

August 2026

- Contributor to an autonomous multi-agent system for alignment research, built with the UK AI Security Institute; evaluation infrastructure, agent isolation, failure-mode analysis of agent-generated output.

MARS Fellowship (Cohort V)

June 2026

- Incoming research fellow under Shivam Raval (Harvard, ICLR/IEEE VIS) and Luiza Corpaci (AMD/TU Wien); investigating scalable oversight via mechanistic detection of semantically unfaithful LLM translations of formal specifications using interpretability techniques.

Algoverse AI Research Program

May 2026

- Investigating Sparse Auto-Encoder (SAE) feature stability from single-run statistics across model family.

EXPERIENCE

Scale AI

San Francisco, USA

AI Engineer — Agentic Capabilities

July 2025 – April 2026

- Building hyper-local deep research agentic systems and automated evaluation framework to benchmark research quality, grounding, and task success across iterations.

Two Sigma

New York, NY, USA

Software Engineer — Platform Engineering

August 2022 – July 2025

- Workflow Orchestration Platform, Distributed systems, Centralized database service, Tech Risk/Authorization/Security.

Amazon

Seattle, WA, USA

Software Development Engineer

April 2020 – July 2022

- Built components of compromised account Sanitization and Cleaning workflow orchestrators.

EDUCATION

University of Southern California

August 2017 – December 2019

Master’s, Computer Science

University of Southern California

August 2013 – May 2017

Bachelor’s, Mechanical Engineering

SKILLS

Python, PyTorch, TransformerLens, Agentic AI systems, RAG, Vector embeddings, Computer Vision, OpenCV, Java, Temporal, AWS, GCP, Kubernetes, Pandas, Linux/Unix, PostgreSQL